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Patient-specific virtual surgical planning for tongue reconstruction: evaluating hyperelastic inverse FEM with four simulated tongue cancer cases

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E-mail: isazadeh@ualberta.ca**Keywords:** plastic and reconstructive surgery, tongue reconstruction, precision medicine, patient-specific virtual surgical planning, soft tissue modeling, image-guided surgery

Abstract

Objective. Anatomically and functionally optimal tongue reconstruction after tumor removal presents significant challenges. Current virtual surgical planning (VSP) utilizes patient-specific data with geometric algorithms for free flap design. However, these geometric approaches often inadequately account for complex soft tissue biomechanics. This study introduces a biomechanics-informed VSP algorithm and computationally compares its flap designs against those derived from purely geometric methods. **Approach.** Hyperelastic inverse finite element method (hiFEM) was developed by integrating an Ogden hyperelastic constitutive model into a predecessor algorithm. The planar flap shape is determined by minimizing potential energy when tissue deforms to match patient-specific MRI-derived 3D defect geometry. Four clinically plausible tongue cancer cases were simulated, and resection regions were delineated. For each case, flap designs were generated using hiFEM, its predecessor iFEM, and two geometric flattening techniques: NURBS surface flattening and boundary first flattening (BFF). Intrinsic tissue deformation for these designs was compared across methods and quantified using area stretch metric. **Main results.** Across all simulated cases, hiFEM-generated flap designs required less intrinsic tissue deformation. Maximum area stretch ranged from 1.10–1.12 for hiFEM designs, versus 1.19–1.38 for NURBS flattening and 1.54–1.74 for BFF designs. Furthermore, hiFEM's area stretch distribution was tighter, centered around one (ideal, no stretch). Geometric comparison showed hiFEM yields flap designs similar to the clinically validated geometric algorithm, NURBS flattening, with an average Hausdorff distance of only 1.3 mm. hiFEM's distinct advantage is its core objective of minimizing tissue stretch, which has clinical relevance and suggests potential for improved patient outcomes. Computationally, hiFEM demonstrated robustness and efficiency. It converged rapidly (8 to 10 iterations; less than 0.3 s/case), even for complex geometries where iFEM failed. **Significance.** hiFEM offers a biomechanically informed and computationally robust tool for tongue VSP, showing potential for broader application in breast, nasal, and other soft tissue reconstructions.

1. Introduction

Glossectomy, the surgical removal of soft tissue, is the primary treatment modality for tongue cancer (Mannelli *et al* 2018, Ansarin *et al* 2019, De Berardinis *et al* 2022). While small resections may permit primary closure, medium-to-large defects necessitate reconstruction to restore critical functions (Mallet *et al* 2009, Ansarin *et al* 2019). Microvascular free flap transplantation represents the preferred reconstructive method whenever feasible (Mallet *et al* 2009). This procedure involves harvesting a two-dimensional (2D) section of skin and subcutaneous tissue, often from the forearm or thigh, and meticulously shaping it to replicate the complex three-dimensional (3D) anatomy of the resected tongue defect (Idris *et al* 2018, Baskin *et al* 2019).

Free flap planning, particularly its volume and planar geometry, directly influences the final structure of the post-surgery tongue (neotongue) and the subsequent restoration of oral functions (Seikaly *et al* 2003, 2009, Urken 2012, Idris *et al* 2018, Baskin *et al* 2019). Achieving optimal functional outcomes demands precise control over not only the flap's bulk but also the geometric configuration of the harvested skin. This configuration must accurately contour the defect's outer surface to restore the tongue's anatomical integrity as closely as possible (Idris *et al* 2018, Baskin *et al* 2019, Fan *et al* 2019a, 2019b). Individualized and defect-specific flap planning demonstrates significant benefits for improving tongue mobility and functional recovery. Studies consistently report improved swallowing (Wang *et al* 2016, Fan *et al* 2019a, 2019b), speech intelligibility (Yi *et al* 2020, Han *et al* 2022, Liu *et al* 2023b), and overall quality of life (QoL) (Liu *et al* 2023b) when the flap planning incorporates the patient's unique anatomical characteristics and defect dimensions.

Effective flap planning, however, faces considerable hurdles. Accurately matching the intricate 3D characteristics of diverse tongue defects to a planar flap template remains a significant challenge. Several studies have explored associations between common tongue defects and specific planar shapes to guide flap planning (Idris *et al* 2018, Baskin *et al* 2019, Fan *et al* 2019a, 2019b, Koumoullis *et al* 2020, Han *et al* 2022). These traditional approaches often involve manual measurements between anatomical landmarks, which are then used to adapt predefined generic templates (Wang *et al* 2016, Fan *et al* 2019a, 2019b, Liu *et al* 2023a). However, substantial variability exists in defect position, size, and spatial geometry, compounded by individual anatomical differences (Engel *et al* 2010, Ansarin *et al* 2019, Koumoullis *et al* 2020, Liu *et al* 2023b). Consequently, these methods can lead to suboptimal flap utilization, intraoperative flap trimming, excessive tensions, and inconsistent functional outcomes, relying heavily on the surgeon's experience and intuition (Fan *et al* 2019b, Liu *et al* 2023b). This underscores the need for innovative, patient-specific virtual surgical planning (VSP) techniques that can provide precise planning and predictable outcomes.

Digital tools and 3D modeling are driving transformative changes in plastic and reconstructive surgery (Logan *et al* 2013, Chuka *et al* 2017, Papadopoulos-Nydam *et al* 2017, Serrano *et al* 2019, Idris *et al* 2022). These technologies can accurately capture a patient's unique anatomy from medical images, such as MRI or CT scans (Fan *et al* 2019a, 2019b, Han *et al* 2022, Xu *et al* 2022, Liu *et al* 2023b). This anatomical data can then inform flap planning, either by adapting predefined templates (Fan *et al* 2019a, 2019b, Han *et al* 2022, Liu *et al* 2023b) or enabling the independent algorithmic design of customized flaps through surface flattening algorithms (Xu *et al* 2022).

An alternative perspective on optimizing flap planning considers the intrinsic deformation when shaping a 2D flap into a 3D geometry. Specifically, designing a flap shape that minimizes the required tissue stretch during this transformation is biomechanically desirable for several reasons. Lower intrinsic stretch implies less internal tension within the shaped flap. This reduction potentially reduces the risks of complications such as compromised blood supply, tissue necrosis, impaired healing, and scar formation (Ogawa *et al* 2012, Tepole *et al* 2014a, 2014b). It also minimizes tethering to surrounding structures, thereby preserving mobility (Urken 2012, Fan *et al* 2019b, Benjamin *et al* 2020, Rui *et al* 2022). Furthermore, minimizing the stored elastic energy should reduce the flap's tendency to recoil after suturing, which allows for better conformity of the flap to the defect. Therefore, designing a flap shape that intrinsically requires less stretch is a key consideration for optimizing reconstructive outcomes.

To achieve this, current VSP tools often employ flattening algorithms. These algorithms map the 3D defect surface onto a 2D plane while aiming to minimize some form of geometric distortion (e.g. changes in area, length, or angle) (Huang *et al* 2007, Xu *et al* 2022). For instance, the Computer-Aided Reconstruction of Soft Tissue (CARST) method, utilizing Non-Uniform Rational B-Splines (NURBS) surface flattening, demonstrated significant clinical advantages (Xu *et al* 2022). Their NURBS-based method flattens a digital representation of the resection surface while attempting to preserve geometric properties during the transformation to create individualized flap-cutting guides. A study involving 30 patients showed significant clinical improvements compared to traditional methods, including reduced flap trimming, shorter operative times, shorter hospital stays, and better defect matching (Xu *et al* 2022). This work validates the substantial value of VSP.

While powerful for geometric translation and computer graphics applications, such methods optimize mapping properties instead of the physical work involved in deformation. Addressing this limitation requires algorithms that accurately account for the biomechanics of soft tissue deformations. Unlike rigid skeletal structures where geometric planning is often sufficient (Logan *et al* 2013, Serrano *et al* 2019), soft tissues undergo complex, nonlinear, large deformations during surgical manipulation. Predictive surgical planning must therefore incorporate realistic models of the tissue behavior. To bridge this gap, we introduce hiFEM. This method builds upon its predecessor—iFEM (Isazadeh *et al* 2024)—but crucially integrates a hyperelastic constitutive model, which is essential for accurately capturing the large, nonlinear mechanical response of the skin under surgical manipulations (Mostafavi Yazdi and Baqersad 2022). The core principle involves finding the optimal planar flap shape by minimizing the deformation energy within the tissue as it deforms to the 3D defect geometry. We hypothesize that optimizing based on the physical deformation work allows hiFEM to yield a flap design that inherently requires less intrinsic stretch to achieve the target shape, compared to designs derived from geometric flattening models.

This study: (1) implements the hiFEM algorithm, incorporating a validated Ogden hyperelastic model, to generate patient-specific flap designs for four simulated tongue reconstruction cases, (2) quantitatively compares the intrinsic flap deformation required to map the planar flap to the target 3D shape for hiFEM designs against those generated by the prior iFEM method (Isazadeh *et al* 2024), NURBS surface flattening (Xu *et al* 2022), and boundary first flattening (BFF) (Sawhney and Crane 2018); (3) assesses the resultant effective stress within hiFEM-designed flaps against relevant physiological thresholds, and (4) evaluates the computational robustness and convergence rate of hiFEM in comparison to the predecessor iFEM algorithm.

2. Method

2.1. Patient anatomy acquisition

Following approval by the Health Research Ethics Board of Alberta (HREBA.CC-17-0597), a male participant in his early 40 s provided written informed consent for magnetic resonance imaging (MRI) of the head and neck region. A Siemens Magnetom Prisma 3 T MRI scanner (Siemens Medical Solutions, Erlangen, Germany) acquired a T1-weighted structural scan of the head and neck. This included the cervical spine, mandible, submental tissues, and palate. The imaging parameters include a 1 mm axial slice thickness, a 0.5 mm in-plane resolution, a 12° flip angle, a repetition time (TR) of 4.72 ms, an echo time (TE) of 2.46 ms, a receiver bandwidth of 650 Hz/pixel, a 256 mm field of view (FOV), and a 512 × 512 matrix size. The resulting images were segmented to produce a 3D model of the tongue using Mimics Medical 17.0 software (Materialise, Leuven, Belgium. 2017).

The 3D model served as the basis for defining four artificially simulated tumor resection cases. These scenarios aligned with Urken's classification (Urken 2012) for class I and II tongue cancers. Figure 1 illustrates the four cases: (A) a hemiglossectomy that leaves the tip of the oral tongue intact, (B) a glossectomy involving half of the oral tongue, (C) a hemiglossectomy extending to the base of tongue posterior side but sparing the root, and (D) an asymmetric partial glossectomy that includes the anterior two-thirds of the oral tongue but extends across the midline. The use of the asymmetric case aims to demonstrate the algorithm's capability in handling complex resections. Two perspectives—top view and isometric view—are shown in figure 1 to aid visualization. The geometric complexity of the required reconstructions increases from case (A) to (D), providing a systematic test of the algorithm's robustness. Preprocessing involves separating the simulated defect from the tongue. The resection surface was then discretized into triangular elements of approximately 0.8 mm size using MeshMixer software (Autodesk Inc. 2018).

2.2. Algorithmic flap design with hiFEM

Virtual free flap planning for tongue reconstruction can be systematically addressed using an inverse approach, introduced as iFEM in our previous work (Isazadeh *et al* 2024). Unlike traditional forward Finite Element Methods that simulate deformation from known initial conditions, our inverse strategy (workflow summarized in figure 2) begins with the desired anatomical outcome: the 3D geometry of the resected tongue defect serves as the target configuration. The algorithm then works backward. It iteratively refines an initial guess to ultimately determine the optimal planar flap shape, defined as the one that will transform into the target 3D configuration with minimal deformation energy. The initial guess is often derived from a geometric flattening technique such as BFF (Sawhney and Crane 2018).

A key challenge in this iterative optimization arises from the soft tissues' inherent material nonlinearity: their mechanical response, and therefore stiffness, depends on the applied deformation. To computationally manage this complexity within the inverse framework, the algorithm employs an iterative scheme centered on the effective secant modulus ($\hat{E}$). Within each iteration, the deformation state for every element must first be determined. This is typically achieved by quantifying the principal stretches required to map that element

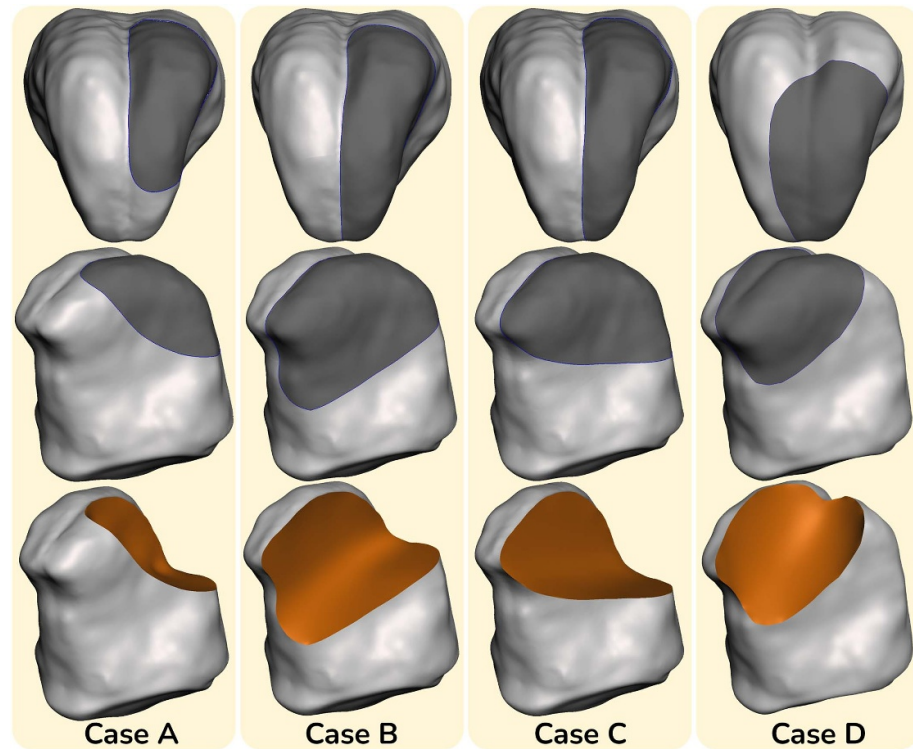


Figure 1. Artificially simulated glossectomy cases. The first and second rows show axial and isometric views with the resected area outlined. The third row shows the isometric view after resecting the outlined region.

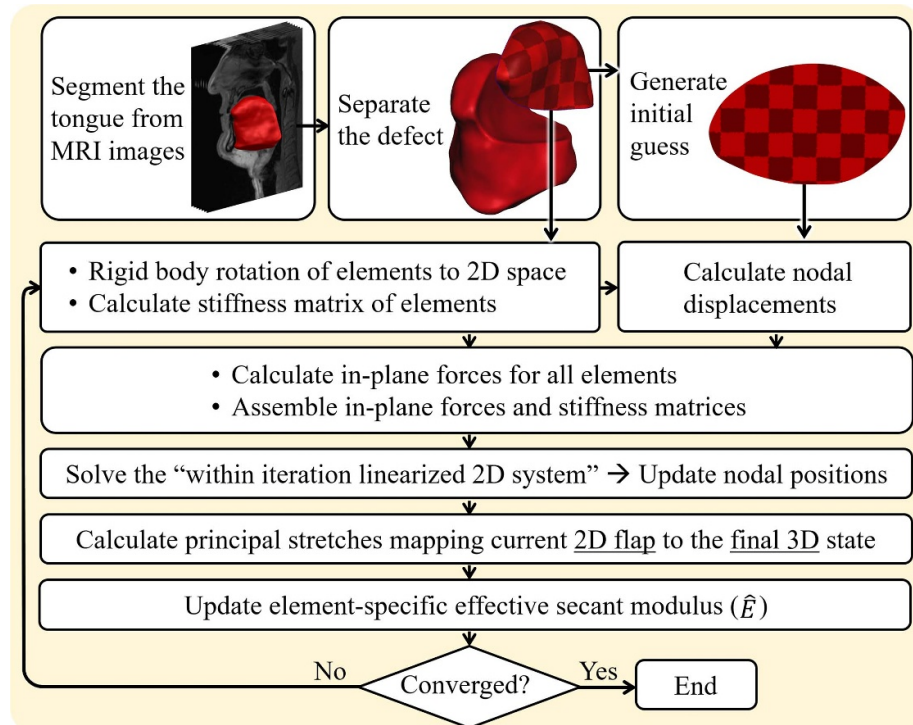


Figure 2. Flowchart summarizing the virtual surgical planning process for tongue reconstruction, from MRI image segmentation to optimal flap design.

from its current 2D configuration to its corresponding state on the fixed 3D target surface. This element-specific deformation is then used for calculating the $\hat{E}$ for that element in the current iteration, based on the chosen nonlinear constitutive model.

The assignment of an element-specific $\hat{E}$ permits a temporary linearization of the problem within each iteration. By utilizing the calculated $\hat{E}$, each element can be treated, for the purpose of that computational step, as if it possessed a constant stiffness equal to its current $\hat{E}$. A global system matrix, incorporating the effective secant stiffness values ($\hat{E}$) of all elements, is assembled under plane stress assumptions in the 2D domain. Solution of this linear system yields nodal displacements, which are used to update the 2D planar shape of the flap geometry. This iterative cycle—calculating element deformations, determining element-specific $\hat{E}$ from the constitutive model, linearizing the step and solving the linearized 2D system, and updating of the planar geometry—is repeated. Convergence occurs when the changes in the calculated $\hat{E}$ across all elements become negligible between successive iterations.

While the inverse approach offers a structured way to optimize flap shapes, the accuracy and robustness, particularly for complex defect geometries, depend on the fidelity of the material model used to determine the effective secant modulus. The earlier implementation (iFEM) utilized simplified models like the power law (Isazadeh *et al* 2024), which may not fully capture the complex, nonlinear stress–strain response of soft tissue under large deformations. This limitation can lead to a failure to converge in challenging cases. To address this, we developed the hyperelastic iFEM (hiFEM). The core advancement in hiFEM is the direct integration of a validated hyperelastic constitutive model into the inverse iterative framework.

Specifically, hiFEM employs the Ogden ($N = 2$) hyperelastic model, chosen because it accurately describes the in-vivo mechanical behavior of human skin across a wide range of deformations (Flynn *et al* 2011). For the material properties, we adopted validated isotropic parameters derived from in-vivo tests on 21 participants' forearm skin (Flynn *et al* 2011) (table 1), a common donor site for tongue reconstruction. Furthermore, we assumed a uniform flap thickness of 2 mm, consistent with clinical observations (Laurent *et al* 2007, Iivarinen *et al* 2011, Groves *et al* 2013, Baskin *et al* 2019).

Within each iteration of the hiFEM, the process begins by quantifying the physical deformation required to map each element from its current 2D configuration to its designated state on the fixed target 3D surface. This kinematic analysis yields the principal stretches ($\lambda_1, \lambda_2, \lambda_3$) for the element. The third principal stretch, λ_3 , perpendicular to the flap surface, is determined by the incompressibility constraint ($\lambda_1 \lambda_2 \lambda_3 = 1$). Once the principal stretches characterizing the element's deformation are known, hiFEM calculates the actual strain energy density stored within that element using the integrated hyperelastic constitutive law. For this work, we employ the Ogden ($N = 2$) model, described by the strain energy function (Holzapfel 2002):

$$W_{\text{Ogden}} = \sum_{i=1}^2 \frac{2\mu_i}{\alpha_i^2} (\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3) - p(J - 1). \quad (1)$$

Here, μ_i and α_i are the Ogden material parameters. The hydrostatic pressure term, $p(J - 1)$, computationally enforces the simplifying assumption of perfect incompressibility ($J = \lambda_1 \lambda_2 \lambda_3 = 1$). Under this common simplification for soft tissues, the strain energy is expressed directly in terms of the total principal stretches (λ_i) (Holzapfel 2002). This calculation directly incorporates the nonlinear, large-deformation mechanical behavior of the skin tissue into the assessment of the deformation energy.

To make the inverse problem computationally tractable using efficient linear solvers, hiFEM derives the element-specific effective secant modulus ($\hat{E}$) required for the iterative linearization step. The value of $\hat{E}$ is determined by relating the calculated hyperelastic strain energy density (W_{Ogden}) to the equivalent energy expression from linear elasticity under incompressible conditions. In principal directions, the strain energy density for a linear elastic, isotropic, and incompressible material ($\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$) subjected to principal strains ($\epsilon_1, \epsilon_2, \epsilon_3$) and Hooke's law ($\sigma_{ij} = \frac{E\nu}{(1+\nu)(1-2\nu)} \epsilon_{kk} \delta_{ij} + \frac{E}{1+\nu} \epsilon_{ij}$) is given by:

$$W_{\text{linear-incomp}} = \frac{1}{2} \sigma_{ij} \epsilon_{ij} = \frac{E}{2(1+\nu)} (\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2) \quad (2)$$

where E, ν, σ_{ij} and ϵ_{ij} are Young's modulus, Poisson's ratio (taken as 0.5 for incompressibility), the components of the Cauchy stress tensor, and the components of linearized strain tensor, respectively. By substituting the hyperelastic energy density (W_{Ogden}) calculated from equation (1) in place of $W_{\text{linear-incomp}}$ in equation (2), and solving for E , we obtain the effective secant modulus $\hat{E}$ for that element in the current iteration:

$$\hat{E} = \frac{2W_{\text{Ogden}}(1+\nu)}{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2} \quad (3)$$

Note that the principal strains ϵ_i in equation (3) correspond to the large deformation state defined by the principal stretches ($\epsilon_i = \ln(\lambda_i)$).

Table 1. Parameters of material models used in hiFEM and iFEM.

Material modeling	Constant parameters
Hyperelastic Ogden model ($N = 2$)	$\mu_1 = 29.159 \text{ kPa}$, $\mu_2 = 2.147 \text{ Pa}$, $\alpha_1 = 1.708$, $\alpha_2 = 40.087$
Power law model	$k_1 = 1000 \text{ MPa}$, $n_1 = 8.802$, $k_2 = 0.0982 \text{ MPa}$, $n_2 = 1.043$

This calculated $\hat{E}$ provides the necessary element-specific secant stiffness for the iterative linearization step. A global system matrix is assembled based on these effective secant stiffness values ($\hat{E}$) for all elements, applying plane stress assumptions within the 2D domain. Solving this linear system yields nodal displacements, which are then used to update the 2D planar shape of the flap geometry. The algorithm iterates through this cycle-calculating stretches, determining W_{Ogden} , deriving $\hat{E}$, solving the linearized system, and updating the 2D geometry-until the changes in the effective secant moduli across all elements become negligible between successive iterations, indicating convergence.

Crucially, because $\hat{E}$ derives directly from a validated hyperelastic model (equation (1)), the overall optimization process genuinely minimizes the physical potential energy associated with the tissue's hyperelastic deformation. This physics-based energy minimization distinguishes hiFEM from methods that rely on minimizing purely geometric distortion metrics or employ less accurate material models. The result is a 2D flap design that is biomechanically optimized to conform to the target 3D defect with minimal intrinsic tissue stretches and stored strain energy.

2.3. Comparative virtual flap design methods

To evaluate how effectively the proposed hiFEM algorithm minimizes intrinsic tissue deformation, we compared its generated flap designs against those from three alternative methods. The first comparison method was the immediate predecessor to hiFEM based on our previous work (Isazadeh *et al* 2024). While iFEM utilizes similar inverse iterative framework to minimize deformation energy, it differs fundamentally by employing a simplified power-law constitutive model to approximate the tissue's stress-stretch response ($\sigma_{eq} = k_1 \epsilon_{eq}^{n_1} + k_2 \epsilon_{eq}^{n_2}$, where σ_{eq} and ϵ_{eq} are the equivalent von Mises stress and strain respectively). For a direct comparison, the power-law parameters (k_1, n_1, k_2, n_2) were determined by fitting this model to the uniaxial stress-stretch data derived from the Ogden hyperelastic model used in hiFEM (table 1). Once the iFEM algorithm converged or reached its iteration limit, the resulting 2D flap geometry was used for subsequent deformation analysis.

The other two comparison methods were representative geometric flattening techniques, reflecting approaches used clinically or established in computer graphics. The first, NURBS surface flattening, represents the methodology of the clinically validated CARST method (Xu *et al* 2022). This technique uses NURBS to represent the 3D defect surface, which is then computationally flattened into a 2D pattern. While the specific flattening function within the commercial CAD software (Rhino, Robert McNeel Associates) used in the CARST method was not detailed, it likely employed the 'Squish' function, designed for non-developable NURBS surfaces, aiming to minimize geometric distortions like changes in length and area. The second geometric method was BFF (Sawhney and Crane 2018), chosen as a state-of-the-art algorithm from computer graphics known for its robustness in producing low-distortion 2D mappings by balancing angle and area preservation. Crucially, both NURBS flattening and BFF optimize the geometric mapping based solely on geometric principles, without incorporating material properties or the physics of tissue deformation.

2.4. Evaluation metrics

The primary metric for comparing hiFEM against alternative flap design algorithms is the intrinsic flap deformation. We quantify the deformation required to shape the planar flap, generated by each of the four methods (hiFEM, iFEM, NURBS surface flattening, BFF), into the target 3D surface. This comparison assesses how effectively each method minimizes the inherent tissue stretch necessary for the reconstruction. For every triangular element within the 2D flap design produced by each method, we first determine the principal stretches, λ_1 and λ_2 , that occur when mapping this planar element to its corresponding state on the target 3D surface. We then calculate the Area Stretch, $J_{\text{area}} = \lambda_1 \lambda_2$, for each element. This metric measures the local change in surface area during the 2D-to-3D transformation; a value of $J_{\text{area}} = 1$ signifies perfect local area preservation, indicating no in-plane stretching or compression, while deviations from unity quantify the magnitude of local deformation. To compare the methods across the four simulated reconstruction cases (A-D), we analyze the distributions of J_{area} using histograms and statistical box plots.

A unique advantage of the physics-based hiFEM approach is its ability to estimate the resulting mechanical stresses within the flap due to the required deformation. Using the converged principal stretches

(λ_1, λ_2 , and $\lambda_3 = 1/J_{\text{area}}$) obtained from the final hiFEM solution, we calculate the Cauchy principal stresses ($\sigma_1, \sigma_2, \sigma_3$) within each element using the integrated Ogden hyperelastic constitutive law (equation (1)). From the principal stresses, we compute the effective (von Mises) stress, σ_{vM} , for each element and report the maximum value across the four cases.

The tertiary metric compares the computational efficiency and robustness of the two inverse methods, hiFEM (using the hyperelastic model) and iFEM (using the power-law model). We assess robustness by evaluating the ability of each algorithm to successfully reach the predefined convergence criterion within a maximum number of 30 iterations. Success or failure to converge is recorded. The convergence rate, indicating the speed at which each algorithm stabilizes, is visualized. This involves calculating the L2 Norm of the vector that contains the relative changes in the effective secant modulus for all elements at each iteration. Specifically, this norm is computed as the square root of the sum of the squared relative changes across all elements ($\sqrt{\sum_{el=1}^n (\Delta \hat{E}_{el} / \hat{E}_{el})^2}$). Faster decay of this norm indicates faster convergence.

Convergence is defined as the point where this L2 Norm falls below a specified threshold (10^{-10}). Reaching this criterion signifies that the iterative updates to the element's effective secant modulus, and consequently the flap geometry, have become negligible, indicating that a stable solution has been found.

The hiFEM and iFEM algorithms were developed in Python, leveraging the NumPy and SciPy libraries for numerical computations. Geometric processing, mesh handling, and visualization tasks were managed using the PyVista library (Sullivan and Kaszynski 2019). To ensure transparency and facilitate reproducibility, the complete source code and all input data files necessary to replicate the findings are publicly available as an open-source package in the hiFEM repository (Isazadeh *et al* 2025). All computational simulations reported in this study were executed on a standard desktop computer (Core™ i5-12400 2.50 GHz with 16 GB installed RAM).

3. Results

The hiFEM successfully generated optimized planar flap designs for all four simulated tongue reconstruction cases (A)–(D). Figure 3 presents the resulting 2D flap geometries obtained from hiFEM alongside those generated by the comparative methods. The figure also includes visualizations mapping the planar flap texture onto the corresponding 3D defect surface for easier orientation.

To quantify the geometric differences between the flap shapes, the maximum Hausdorff distance was calculated between the boundary of the hiFEM-designed flap and the boundaries of the flaps from the other methods (table 2). The geometric difference between hiFEM and iFEM flaps was negligible (0.09–0.14 mm) for the cases (A)–(C); iFEM failed to converge for the complex asymmetric Case (D). Larger geometric deviations were observed when comparing hiFEM to the geometric flattening techniques.

Figures 4 and 5 focus on quantifying the intrinsic tissue deformation, measured by the Area Stretch ($J_{\text{area}} = \lambda_1 \lambda_2$), required to map the 2D planar flap designs onto the target 3D defect surfaces for each of the four reconstruction cases (A)–(D). They present comparative histograms and statistical box plots illustrating the distributions of J_{area} for flaps designed using hiFEM, iFEM, NURBS surface flattening, and BFF.

A consistent pattern emerged across all four simulated cases: flap designs generated by hiFEM required less intrinsic deformation compared to those produced by the geometric flattening techniques. Specifically, the maximum observed area stretches for hiFEM designs ranged from 1.10 to 1.12. In contrast, maximum J_{area} values were higher for NURBS flattening, ranging from 1.19 to 1.38, and substantially higher for BFF, ranging from 1.54 to 1.74. The average J_{area} values showed a similar trend, indicating that hiFEM designs necessitate lower overall stretch to achieve the target 3D geometry. Furthermore, the distribution of J_{area} values for hiFEM designs was generally tighter and more concentrated around the ideal value of 1.0 (no area change), suggesting a more uniform and less severe deformation pattern compared to the geometric methods (figures 4 and 5). While hiFEM and iFEM yielded similar J_{area} distributions for Cases (A)–(C), hiFEM proved more robust. It successfully converged for all four cases, including the complex Case (D), whereas iFEM failed to converge for Cases (B) and (C) and diverged for Case (D). This demonstrates the hyperelastic model's enhanced capability to handle challenging deformations.

hiFEM's physics-based nature uniquely enables estimation of mechanical stress (σ_{vM}) within the flap. The analysis revealed low resultant stress levels for all flap designs generated by hiFEM. The maximum effective stresses (σ_{vM}) calculated for Cases (A)–(D) were 18.0 kPa, 21.5 kPa, 21.3 kPa, and 24.5 kPa, respectively.

Another critical check enabled by the physics-based hiFEM is the verification of material volume preservation, consistent with the near-incompressible nature of skin tissue (Mostafavi Yazdi and Baqersad 2022). The underlying hyperelastic model within hiFEM enforces this constraint. For all four cases, the volume change between the 2D planar flap and the deformed flap was less than 10^{-5} percent, confirming the algorithm's adherence to the incompressibility constraint.

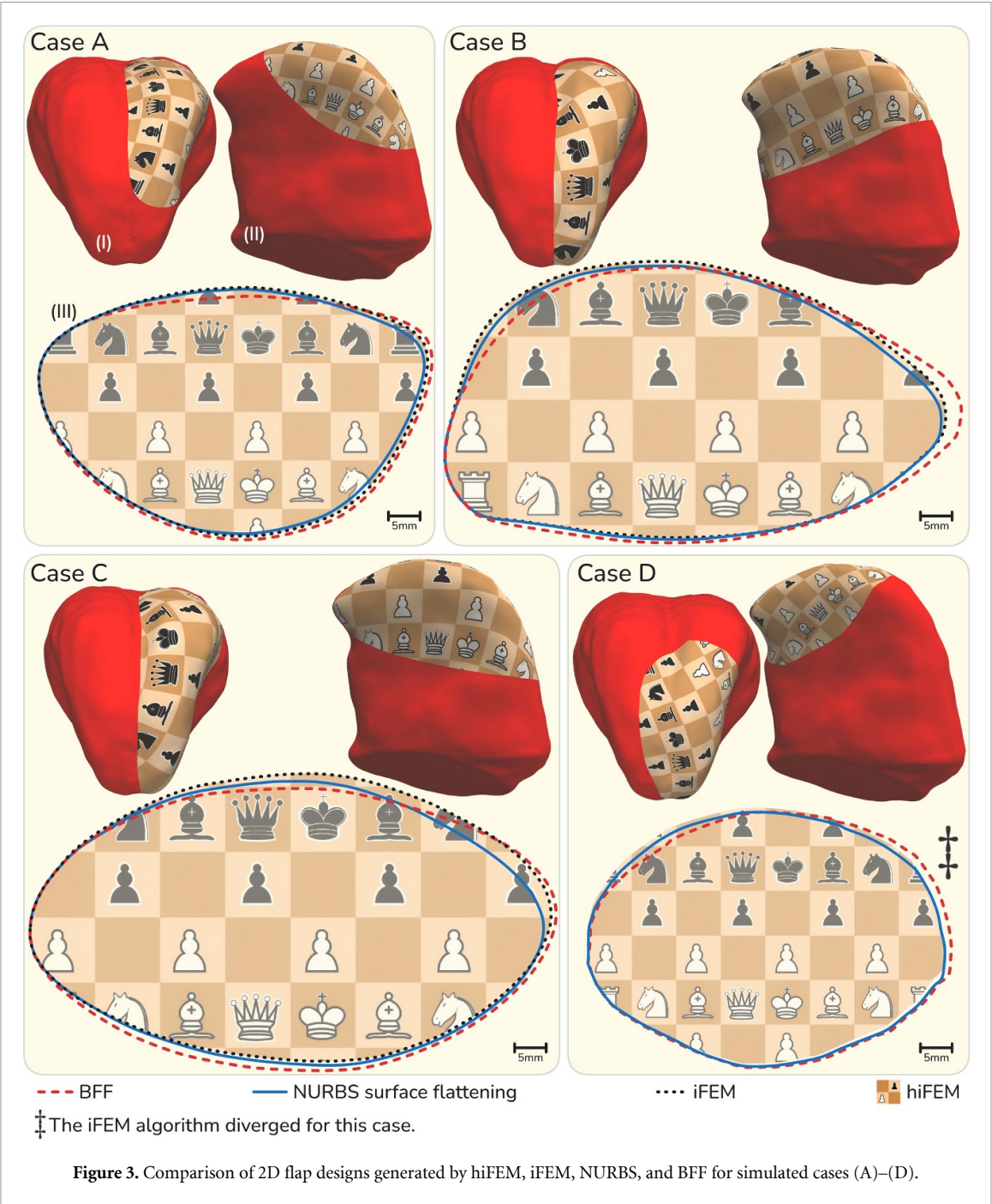
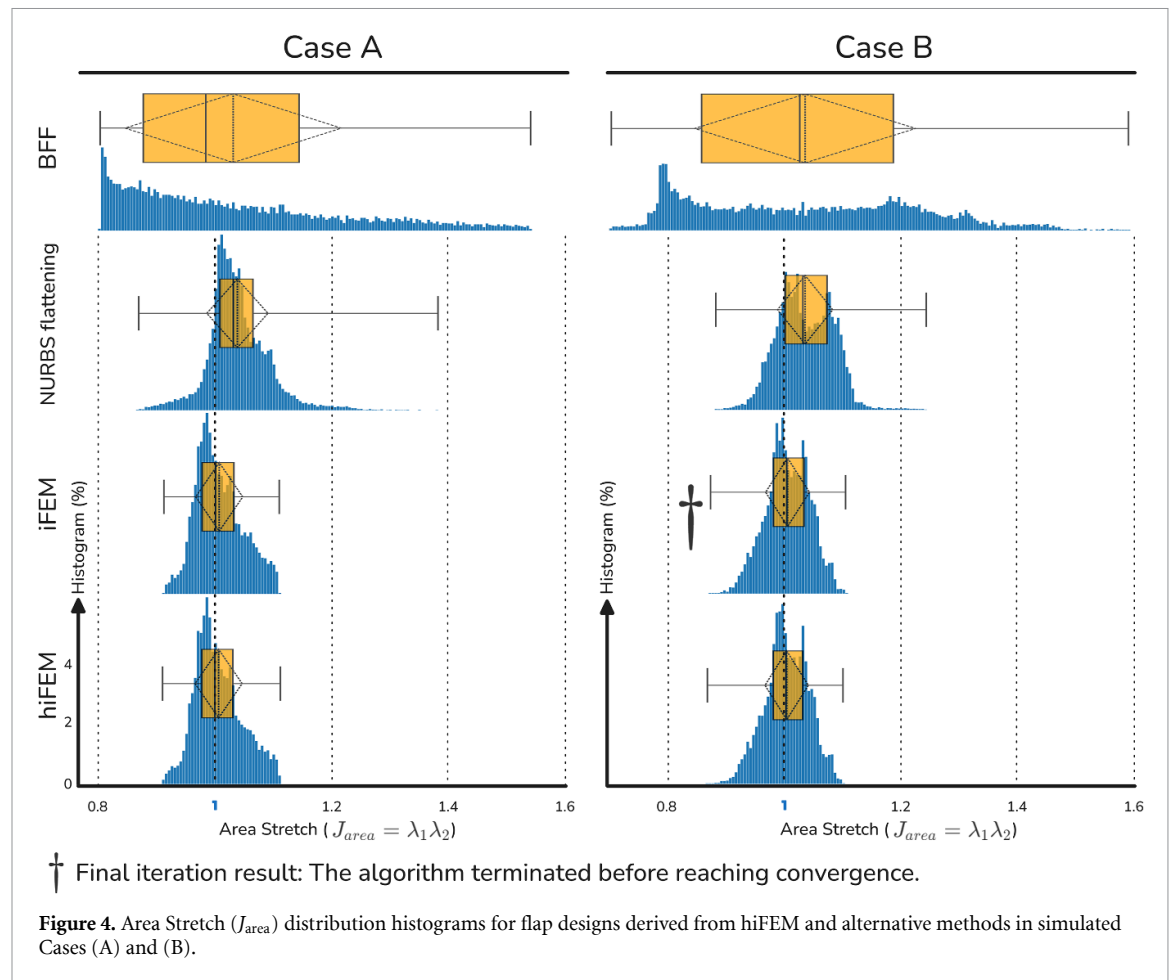


Table 2. Maximum Hausdorff distance comparing flap boundary geometries generated by hiFEM versus alternative methods for four reconstruction cases.

Case	hiFEM vs iFEM (mm)	hiFEM vs NURBS flattening (mm)	hiFEM vs BFF (mm)
A	0.09	0.90	1.50
B	0.12	1.20	2.40
C	0.14	2.30	2.40
D	—	0.90	2.00

Lastly, computational performance of hiFEM was compared to the prior iFEM algorithm. Figure 6 illustrates the convergence behavior by plotting the L2 Norm of relative changes in effective secant moduli against iteration count. hiFEM consistently reached the convergence criterion ($L2 \text{ Norm} < 10^{-10}$) within 8–10 iterations for all four simulated cases (A)–(D). In contrast, iFEM failed to converge within 30 iterations for Cases (B) and (C) and diverged for Case (D). On the standard desktop hardware, hiFEM completed the optimal flap design calculation in under 0.3 seconds per case.



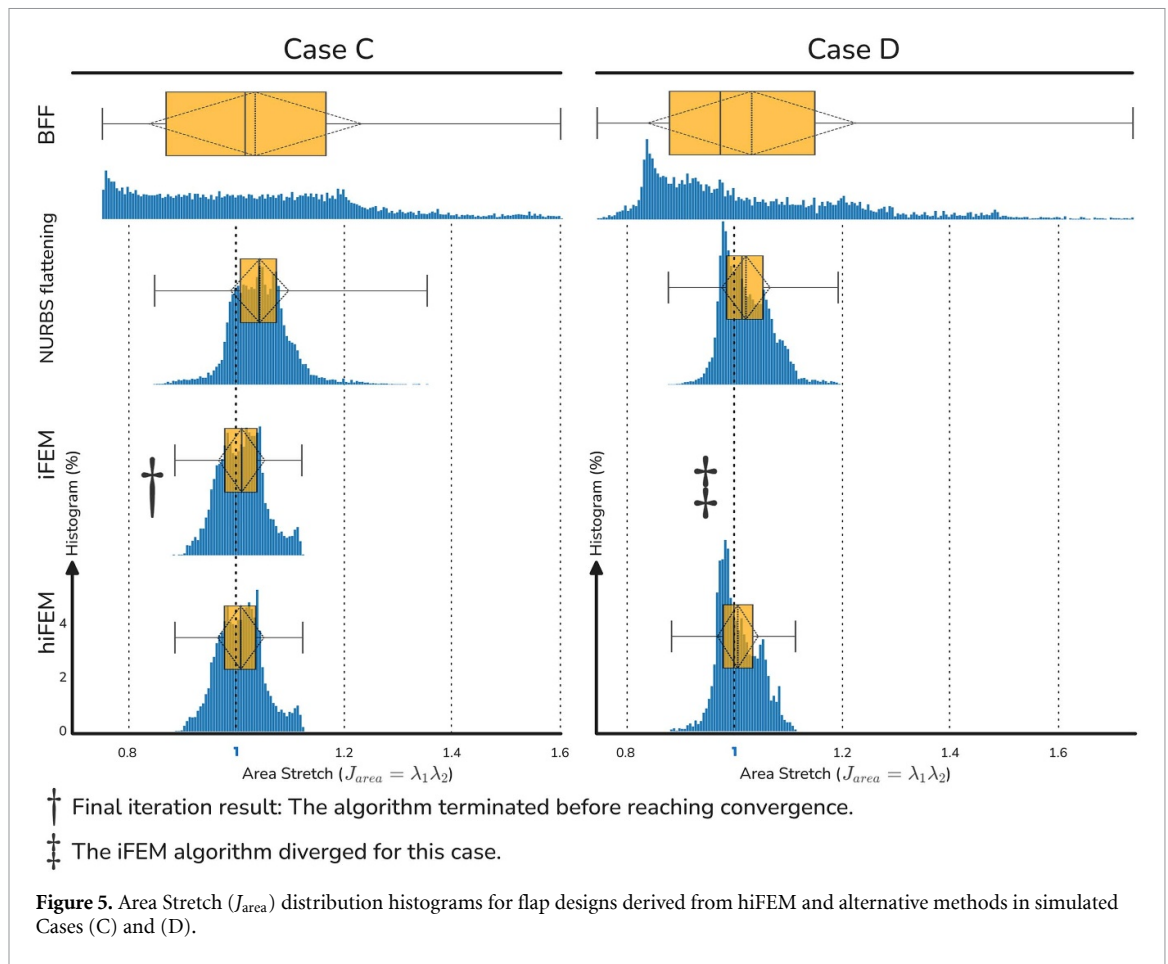
4. Discussion

This study introduces hiFEM, a digital tool for algorithmic free flap planning in soft tissue reconstructive surgeries. It advances an earlier algorithm by integrating Ogden hyperelastic model consistent with human skin's mechanical properties (Flynn *et al* 2011, Mostafavi Yazdi and Baqersad 2022). Consequently, hiFEM generates biomechanically optimized flap designs by accurately simulating large tissue deformations.

Unlike traditional patient-specific approaches that rely on manual measurement with instruments like flexible rulers to inform and adapt flap templates (Wang *et al* 2016, Idris *et al* 2018, Baskin *et al* 2019, Liu *et al* 2023a), hiFEM leverages MRI data for virtual surgical planning. MRI-guided models provide detailed anatomical information and improve the precision of the flap planning process. Past studies have shown that MRI-based, patient-specific models enhance surgical precision and functional outcomes (Fan *et al* 2019a, 2019b, Han *et al* 2022, Xu *et al* 2022, Liu *et al* 2023b).

However, a digital tongue model alone is insufficient for VSP; determining the corresponding planar free flap geometry is crucial. Current methods like ICTM (Liu *et al* 2023b) and FILEPS (Fan *et al* 2019a, 2019b, Han *et al* 2022) achieve this by manually identifying anatomical landmarks on MRI scans, measuring distances between them, and using these measurements to guide predefined flap templates based on resection type. While systematic, these approaches require significant surgical expertise, particularly for selecting the appropriate template and accurately translating complex 3D anatomy into a 2D design. In contrast, hiFEM automates this critical translation using an inverse, physics-informed algorithm. By incorporating hyperelastic skin behavior, hiFEM produces flaps that are anatomically tailored and mechanically optimized for the specific resection.

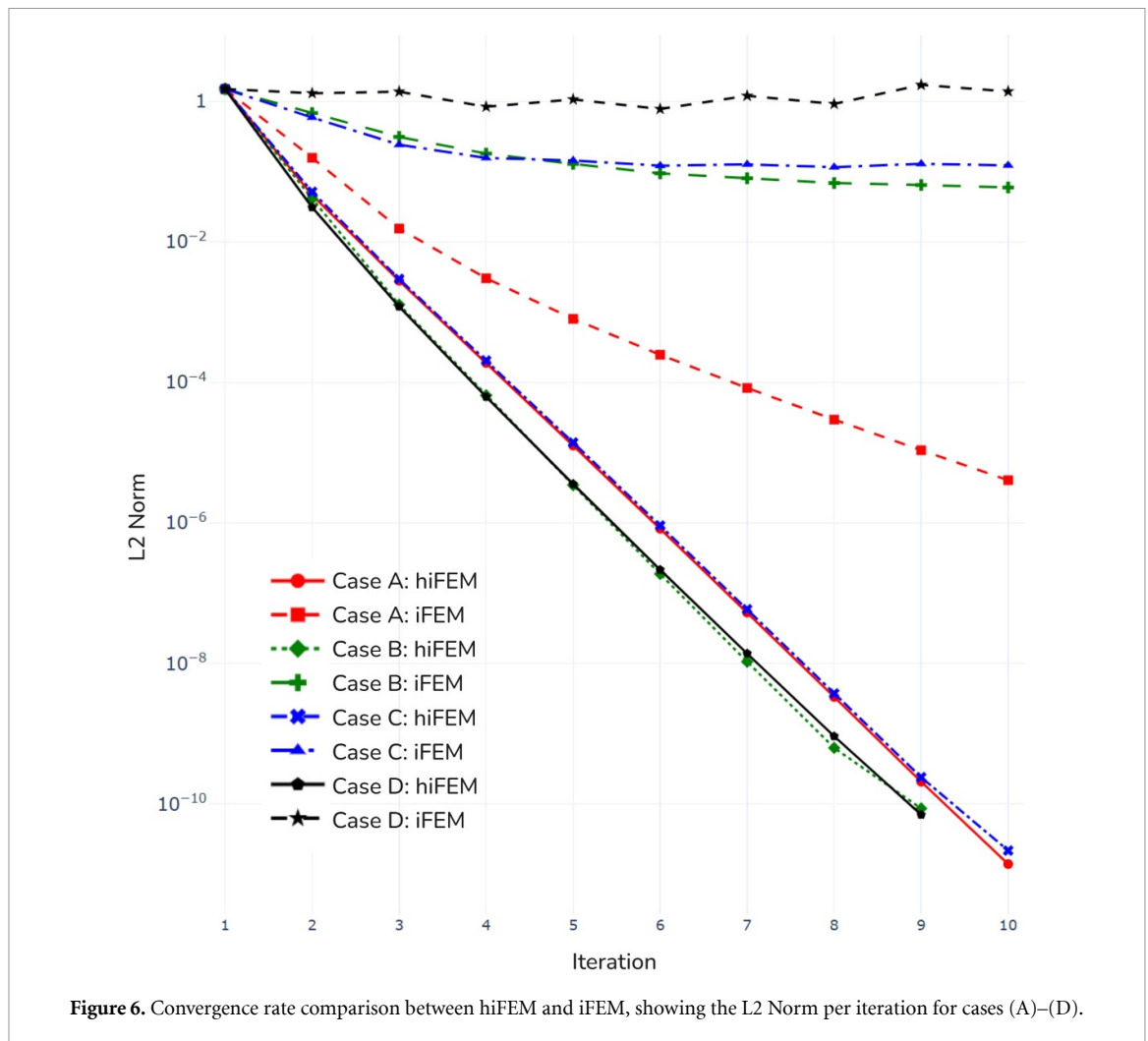
While methods like CARST (Xu *et al* 2022) represent a significant, clinically validated improvement over traditional techniques by leveraging patients' imaging data, digital modeling of their tongue, and computer-aided individualized flap planning, hiFEM provides incremental refinement to this approach through its different optimization objective. Rather than minimizing purely geometric distortions, hiFEM optimizes the flap design based on the underlying mechanics of soft tissue deformation; it focuses on



designing flap shapes that require less internal tissue stretching when being shaped into the desired 3D geometry.

This optimization objective is clinically relevant because excessive tensile stretch ($\lambda > 1$) leads to high tissue tension and stored energy, which can have immediate adverse consequences. High tension can impede blood flow within the delicate flap vasculature, increasing the risk of ischemia, partial necrosis, and impaired healing (Tepole *et al* 2014a, 2014b, Rui *et al* 2022). Studies also suggest a correlation between mechanical tension, particularly near suture lines, and the development of keloids and hypertrophic scarring (Ogawa *et al* 2012, Van De Water *et al* 2013). Furthermore, excessive tension can pull on surrounding native tongue tissue, restricting the mobility crucial for speech and swallowing (Urken 2012, Fan *et al* 2019b, Benjamin *et al* 2020, Rui *et al* 2022, Liu *et al* 2023b). Maintaining a near-normal neotongue structure, facilitated by appropriate flap design, is known to reduce tethering and allow maximum mobility of the tongue remnant (Urken and Biller 1994, Urken 2012).

Beyond these immediate effects, the stored elastic energy within a stretched flap influences post-suturing outcomes, particularly regarding the conformity of the flap to the defect. After suturing, the flap tissue naturally seeks to release this stored energy and recoil toward its unstretched state. The magnitude of this recoil is directly related to the initial stretch and the tissue's stiffness. A common challenge in tongue reconstruction is the stiffness mismatch, where the harvested skin flap is often considerably stiffer than the underlying native tongue muscle or adipose tissue graft bed (Buchaillard *et al* 2007). Consequently, the recoil forces exerted by a suboptimally designed (highly stretched) flap can disproportionately deform the softer surrounding tissues. Instead of the flap conforming precisely to the defect contours, its recoil pulls the neotongue structure away from the intended anatomical shape, leading to deviations from the surgical plan and potentially compromising function. Therefore, designing a flap shape that intrinsically requires less stretch is crucial not only for immediate tissue viability, improved healing, and minimizing tethering but also essential for achieving accurate anatomical reconstruction. Lower, more uniform stretches promote better, more natural conformity of the flap to the defect shape with less adverse deformation of the neotongue



structure post-surgery. By minimizing the hyperelastic potential energy, hiFEM directly addresses these key biomechanical factors.

Comparing the intrinsic deformation (J_{area}) required by hiFEM flap designs to alternatives highlights the benefit of its physics-based optimization. Across all four simulated cases, hiFEM consistently designed flaps requiring less intrinsic area stretch (lower maximum J_{area} values and distributions tighter around 1.0) compared to established geometric flattening methods (figures 4 and 5). This superiority stems from hiFEM's optimization objective.

Regarding the comparison with NURBS-based flattening, it is crucial to first acknowledge the significant clinical success and validation of the CARST approach presented by Xu *et al* (2022). Their work clearly demonstrated the substantial value of computer-aided surgical planning over traditional, non-image-guided methods, showing measurable improvements in operative time, flap utilization, defect matching, and functional outcomes. While NURBS flattening represents a proven advance in leveraging patient-specific geometry, hiFEM offers a potential next step in refinement by optimizing based on tissue biomechanics rather than purely geometric distortion metrics.

While hiFEM produces flap shapes geometrically distinct from those generated by NURBS flattening (0.9–2.3 mm maximum Hausdorff distance, table 2), achieving such fine precision consistently during surgical resection and flap harvesting remains practically challenging. Therefore, the full realization of the clinical benefit derived purely from this subtle shape difference might be influenced by the accuracy of surgical execution. However, the rationale for optimizing the design itself remains compelling. Surgical planning and execution involve multiple stages where inaccuracies can accumulate. By starting with a flap design that is biomechanically optimized, we minimize a known source of potential suboptimality inherent in the plan itself. Minimizing error at the design stage is crucial to avoid compounding inaccuracies that could negatively impact the reconstruction. Thus, while NURBS surface flattening signifies a major, clinically validated improvement, hiFEM's physics-based approach offers a theoretically sound refinement aimed at

further enhancing biomechanical outcomes. Additionally, it is worth noting that while the specific NURBS flattening tool used in the CARST study is part of commercial software (Rhino 3D), our hiFEM implementation, along with documentation and application examples, is available as an open-source package (Isazadeh *et al* 2025).

Computationally, hiFEM outperforms its predecessor (Isazadeh *et al* 2024), demonstrating enhanced robustness by converging reliably across all cases, even complex geometries where iFEM failed. The key advantage stems from hiFEM's use of the Ogden hyperelastic model. This model not only represents skin's large deformations more accurately than the previous power-law approach but also features a well-behaved strain energy function. This property likely improves the numerical stability of the iterative secant modulus updates within the inverse algorithm. With optimal designs generated in under 0.3 s, hiFEM achieves practical clinical feasibility. It offers an efficient, direct optimization, which avoids the prohibitive computational cost and instability associated with trial-and-error forward simulation approaches (Tepole *et al* 2014b, Anderson *et al* 2017). Consequently, hiFEM's speed and reliability position it as a promising tool for VSP integration.

Lastly, the physics-based nature of hiFEM uniquely allows for the estimation of the resultant stresses within the flap after conforming to the defect shape. This provides an additional layer of design evaluation unavailable from purely geometric approaches. The hiFEM stress estimations for all designs are considerably below thresholds associated with compromised tissue viability—around 1 MPa (Tepole *et al* 2014a). They are also lower than physiological stresses during functional activities like swallowing—around 100 kPa (Hermant *et al* 2017). These estimations, upon proper future validation studies, could provide independent, physics-based evidence corroborating the primary finding of minimized intrinsic deformation. This further supports the biomechanical soundness of hiFEM designs and their potential to reduce tension-related complications.

Despite its promise, this study has limitations that guide future research directions. The current work was based on MRI data from a single healthy participant; validation across a diverse patient cohort, encompassing a range of anatomies and demographics, as well as a broader spectrum of defect types—including complex geometries involving the floor of the mouth or those requiring total/subtotal reconstructions—is a crucial next step. Furthermore, the current implementation employs validated isotropic hyperelastic parameters (Flynn *et al* 2011), a simplification that neglects known biomechanical complexities such as anisotropy, pre-stress, and the significant variations in skin properties due to anatomical donor site, age, and sex. Therefore, a critical next step is to perform a sensitivity analysis to quantify how material property variations impact the optimized flap design. Ultimately, the long-term vision for VSP in tongue reconstruction involves incorporating patient-specific material properties—an enhancement uniquely achievable with physics-based methods like hiFEM.

Beyond refining the material model and chosen parameters, future computational studies must account for the mechanical interplay between the flap and the recipient site. Such simulations would be essential to predict the final post-suturing anatomical conformity. These predictive models could eventually be extended to incorporate long-term biological processes such as flap atrophy (Gewirtz *et al* 2024) and radiation-induced tissue fibrosis (Al-Zanoon *et al* 2024) as well. Crucially, the computational studies must be followed by benchtop experiments and prospective clinical studies. These are essential to rigorously evaluate the real-world efficacy of hiFEM-guided reconstructions against standard-of-care techniques. This future work must also address the critical challenge of translating the virtual plan to the operating room, particularly given the tongue's mobility, by developing and validating patient-specific surgical guides that ensure accurate registration of the planned resection onto the intraoperative anatomy.

A critical consideration for clinical translation is the placement of the vascular pedicle. The hiFEM algorithm accommodates this surgical requirement within an integrated VSP workflow. First, the surgeon identifies the optimal pedicle location on the 3D defect model. The algorithm's deterministic 3D-to-2D mapping then precisely transfers this point to the biomechanically optimized 2D flap template. This marked template guides the surgeon during harvest to align the flap with the donor vessel. In this manner, hiFEM integrates, rather than automates, the surgeon's crucial anatomical judgment; it provides a biomechanically sound design that respects clinical constraints. Future work could also incorporate pedicle length and orientation as explicit optimization constraints.

The fundamental principle of hiFEM renders it a versatile tool with broad applicability for VSP across various soft tissue reconstruction domains. It offers potential biomechanical advantages in applications such as breast reconstruction (Huang *et al* 2007), nasal reconstruction (Wei *et al* 2022), and the planning of complex wound closures (Ji *et al* 2024), where minimizing tissue tension and ensuring conformational accuracy are similarly crucial. The capacity to integrate patient-specific anatomy and biomechanics provides a pathway to enhance precision and predictability in these challenging reconstructive scenarios, moving beyond purely geometric planning towards physics-informed optimization.

5. Conclusion

This study introduced hiFEM, a VSP tool that incorporates tissue biomechanics into algorithmic free flap design. Our comparative analysis across four simulated reconstruction cases demonstrated that hiFEM-generated flaps require less intrinsic tissue deformation compared to designs derived from a clinically validated flattening algorithm. In theory, this algorithmic improvement carries clinical relevance. Reduced internal stretches may improve tissue viability and lower tissue tethering to surrounding structures, thereby enhancing mobility. Furthermore, the minimization of stored elastic energy can mitigate flap recoil, which promotes better anatomical conformity of the flap to the defect. Optimizing for minimal tissue stretches, hiFEM offers a physics-informed VSP method yielding biomechanically sound flap designs.

Beyond the biomechanical advantages, hiFEM demonstrated superior robustness compared to its predecessor (iFEM). Its rapid convergence and enhanced robustness, even for complex cases where the previous iFEM failed, establish its practical feasibility for integration into clinical VSP workflows. This holds promise for improving the predictability and functional effectiveness not only of tongue reconstruction but potentially extending to other challenging soft tissue procedures, such as breast reconstruction, nasal reconstruction, and complex wound closure. Ultimately, this method aims to enhance patient-specific surgical precision.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/is-amir/hifem>.

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Conflict of interest

The authors declare that they have no conflict of interest.

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